

Backend Project Setup Checklist (Windows)

Node.js / Express / MongoDB / JWT / Arcjet / Upstash / Nodemailer

1. Install Core Tools

- ☐ Node.js (LTS) — nodejs.org — verify: `node -v` / `npm -v`
- ☐ Git for Windows — git-scm.com
- ☐ VS Code (or WebStorm)
- ☐ Postman or Thunder Client (API testing)
- ☐ MongoDB Compass (optional GUI)

2. Create Free Accounts / Services

- ☐ MongoDB Atlas — cluster + DB user + IP whitelist + connection string
- ☐ Upstash — for QStash / Upstash Workflow
- ☐ Arcjet — get API key
- ☐ Gmail — enable 2FA → generate App Password (for Nodemailer)
- ☐ Hostinger or other VPS — only needed at deployment stage

3. Initialize Project

- ☐ `mkdir project-name && cd project-name`
- ☐ `npm init -y`
- ☐ Set `"type": "module"` in `package.json`
- ☐ Add scripts: `"dev": "nodemon app.js", "start": "node app.js"`

4. Install Dependencies

- ☐ `npm install express mongoose dotenv jsonwebtoken bcryptjs cookie-parser @arcjet/node @upstash/workflow nodemailer dayjs`
- ☐ `npm install -D nodemon`

5. Environment Variables (.env files)

- ☐ Create `.env.development.local` (and `.env.production.local` later)
- ☐ Add: `PORT`, `NODE_ENV`, `DB_URI`, `JWT_SECRET`, `JWT_EXPIRES_IN`
- ☐ Add: `ARCJET_KEY`, `ARCJET_ENV`
- ☐ Add: `QSTASH_TOKEN`, `QSTASH_URL`
- ☐ Add: `EMAIL_PASSWORD` (Gmail App Password)
- ☐ Add `.env*` to `.gitignore`

6. Verify Before Coding

- ☐ `node -v` and `npm -v` return version numbers
- ☐ `git --version` returns a version number
- ☐ MongoDB Atlas connection string tested (e.g., via Compass)
- ☐ Postman/Thunder Client installed and ready to hit localhost